and the time-reversal symmetry

$$\mathscr{F}_t \circ \boldsymbol{R} \circ \mathscr{F}_t = \boldsymbol{R}$$

have a formal similarity, but they are essentially unrelated. A method can be symplectic but not time-reversible (e.g. Symplectic Euler) or it can be time-reversible and not symplectic (e.g. Trapezoidal Rule). However there are intriguing relationships between these properties which we discuss here.

Interestingly the time-reversal symmetry provides analogous properties in some respects to the properties of Hamiltonian systems. For example, consider a linear Hamiltonian system $\mathrm{d}\boldsymbol{z}/\mathrm{d}t = \boldsymbol{JAz}$, with $\boldsymbol{A}$ a symmetric matrix. If λ is an eigenvalue of $\boldsymbol{JA}$ then $\boldsymbol{JAu} = \lambda\boldsymbol{u}$ for some eigenvector $\boldsymbol{u} \neq 0$. Because the matrix $\boldsymbol{JA}$ is real, we know that $\bar{\lambda}$ will also be an eigenvalue. At the same time, we know that since λ is an eigenvalue of $\boldsymbol{JA}$ it is also an eigenvalue of its transpose $(\boldsymbol{JA})^T = \boldsymbol{A}^T\boldsymbol{J}^T = -\boldsymbol{AJ}$, thus

$$-\boldsymbol{AJu} = \lambda\boldsymbol{u}$$

multiplying by $\boldsymbol{J}$ and setting $\boldsymbol{v} = \boldsymbol{Ju}$ we have

$$-\boldsymbol{JAv} = \lambda\boldsymbol{v}$$

implying that $-\lambda$ (and hence also $-\bar{\lambda}$) is an eigenvalue of $\boldsymbol{JA}$. Real eigenvalues of $\boldsymbol{JA}$ are paired with their negatives. If the imaginary part is nonzero, the eigenvalues occur in quadruplets $\{\pm\lambda, \pm\bar{\lambda}\}$. The flow map is

$$\mathscr{F}_t(\boldsymbol{z}) = e^{t\boldsymbol{JA}}\boldsymbol{z}$$

and the exponential matrix will inherit a related structure within the spectrum: λ an eigenvalue of $\exp(t\boldsymbol{JA})$ implies that $\bar{\lambda}$, $1/\lambda$ and $1/\bar{\lambda}$ are all eigenvalues of $\exp(t\boldsymbol{JA})$. This eigenvalue structure is generic for linear symplectic maps in general.

Suppose now we have a linear time-reversible map $\Phi(\boldsymbol{z}) = \boldsymbol{Tz}$, then

$$\boldsymbol{T}^{-1} = \boldsymbol{RTR}.$$

Given an eigenvalue, eigenvector pair $(\lambda, \boldsymbol{u})$ of $\boldsymbol{T}$, let $\boldsymbol{u} = \boldsymbol{Rv}$, so that $\boldsymbol{Ru} = \boldsymbol{R}^2\boldsymbol{v} = \boldsymbol{v}$, then

$$\boldsymbol{T}^{-1}\boldsymbol{v} = \boldsymbol{RTRv} = \boldsymbol{RTu} = \lambda\boldsymbol{Ru} = \lambda\boldsymbol{v}.$$

Thus λ is an eigenvalue of $\boldsymbol{T}^{-1}$ which, in turn, implies that $1/\lambda$ is an eigenvalue of $\boldsymbol{T}$. The matrix being real implies that the conjugates of λ and $1/\lambda$ are also eigenvalues, thus we have the same eigenvalue quadruplets as for a linear symplectic map.

Recall that a pair of maps $\boldsymbol{\Phi}$ and $\boldsymbol{\Psi}$ are said to be conjugate if there is a homomorphism $\boldsymbol{\chi}$ such that

$$\boldsymbol{\Phi} = \boldsymbol{\chi}^{-1}\boldsymbol{\Psi}\,\boldsymbol{\chi}.$$

In such a case the iterates of the two maps will also be conjugate and, if they are numerical methods, they will have similar stability properties and performance (e.g. the same effective order). It is difficult to separate the relevance of the two properties in cases where symplectic and reversible maps are conjugate. This is however rarely the case and certainly does not hold generically for discrete maps in many dimensions [209]. There is no direct correspondence between reversible and symplectic maps, however each class of maps admits certain theorems of dynamical systems which are in many ways analogous (for example, the Kolmogorov-Arnold-Moser, or KAM, theory for symplectic maps near elliptic fixed points [386] has an analogue for reversible maps [97]).

Of particular importance for molecular dynamics are the following properties: a symplectic map will preserve volume, whereas a time-reversible map need not do so, and a symplectic integrator will approximately conserve energy due to the existence of the perturbed Hamiltonian, whereas a time-reversible integrator may give rise to a drift in energy [166].

3.7 Molecular Dynamics with Hard-Sphere Collisions

We close this chapter with a discussion of molecular dynamics with hard-core repulsion. This is a system which is not, properly speaking, Hamiltonian, but which has dynamics that are largely defined by a Hamiltonian system. The added complication is the nonsmooth character of the motion as impulses are applied at points of collision. The discussion below is a summary of work in [40, 184].

We assume, as usual a Hamiltonian $H = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$ but we add the inequality constraint $\|\boldsymbol{q}_i - \boldsymbol{q}_j\| \geq \sigma_i + \sigma_j$ where σ_i, $i = 1, \ldots, \mathrm{N}$ is a core radius. The condition $\|\boldsymbol{q}_i - \boldsymbol{q}_j\| = \sigma_i + \sigma_j$, some i, j defines the constraint surface. When the particles are not touching, they move along Newtonian paths defined by the standard equations of motion. At impact, they exchange momentum and energy according to the rules of elastic collision. Specifically, at the point of contact, the momentum vectors of the two spheres are adjusted according to the rule:

$$\boldsymbol{p} := \boldsymbol{p} + \alpha\boldsymbol{u}_\perp$$

where $\boldsymbol{u}_\perp$ is normal to the constraint surface and α is a parameter chosen to maintain the conservation of energy. Let $\mathscr{R}_\mathrm{c}$ denote the action of the collision operator on the vector of positions and momenta. Then we can write the evolution formally as

$$\mathscr{F}_\tau^{\mathrm{h.s.}}(\boldsymbol{q},\boldsymbol{p}) = \mathscr{G}_{\Delta\tau_r} \circ \mathscr{R}_\mathrm{c} \circ \mathscr{G}_{\Delta\tau_{r-1}} \ldots \circ \mathscr{R}_\mathrm{c} \circ \mathscr{G}_{\Delta\tau_0}$$